HARDWARE ANALYSIS & BILL OF MATERIALS

RoboFace — ESP32 OLED Robot Face

A WiFi-connected ESP32 driving a 0.96" OLED that mirrors the RoboFace web app over Firebase, with over-the-air firmware updates.

Generated from project firmware (`firmware/roboface_oled.ino`) & app. Revision A.

1. What the board must do

REQUIREMENT	SOURCE IN PROJECT	IMPLICATION FOR THE BOARD
Drive a 0.96" SSD1306 128×64 OLED over I²C	Adafruit_SSD1306, addr `0x3C`	I²C bus (SDA/SCL) + 3.3 V to the panel
I²C pins SDA=GPIO21, SCL=GPIO22	`Wire.begin()` defaults	Route these two nets to the OLED header
WiFi (Firebase sync)	WiFi.h + Firebase client	ESP32 with PCB/onboard antenna; clean RF area
OTA self-update	HTTPUpdate, dual-app partition	**≥ 4 MB flash** (min_spiffs = 1.9 MB/app)
Programming + power	USB upload / serial @115200	USB 5 V → 3.3 V regulator + USB-UART (or dev board)

> **Two build paths.** Path A — Module build fastest, no fine-pitch soldering, great for 1–5 units. Path B — Custom PCB integrated single board for a clean product / batch.

2. Path A — Module build (recommended to start) ★

An ESP32 dev board already integrates the 3.3 V regulator, USB-UART and auto-reset, so you only add the display.

#	COMPONENT	SPEC / PART	QTY	PURPOSE
1	ESP32 dev board	ESP32-DevKitC / NodeMCU-32S (ESP32-WROOM-32, 4 MB)	1	MCU + WiFi + USB + power + programmer
2	OLED display	0.96" SSD1306 128×64, I²C (NFP1315)	1	The robot face
3	Jumper wires	Female–female Dupont	4	VCC, GND, SCL, SDA
4	USB cable	Micro-USB / USB-C (matches board)	1	Power + flashing
5	(optional) Carrier/perfboard or 3D-printed mount	—	1	Hold board + OLED together

Wiring (both paths)

OLED PIN	ESP32 PIN	NET
VCC	3V3	+3.3 V
GND	GND	Ground
SCL	GPIO22	I²C clock
SDA	GPIO21	I²C data

```
ESP32 DevKit  (USB = 5V + flashing)      OLED  SSD1306 0x3C
     3V3 ●──────────────────────────────● VCC
     GND ●──────────────────────────────● GND
  GPIO22 ●──────────────────────────────● SCL
  GPIO21 ●──────────────────────────────● SDA
```

3. Path B — Custom single-board PCB (ESP32-WROOM-32E)

Everything the dev board hides, made explicit. Reference designators are suggestions for your schematic.

3.1 Core

REF	COMPONENT	SPEC / PART	QTY	PURPOSE
U1	ESP32 module	ESP32-WROOM-32E-N4 (4 MB) or -N8 (8 MB)	1	MCU, WiFi, PCB antenna, flash
DS1	OLED panel/module	SSD1306 128×64 I^2C (or SH1106 1.3")	1	Display

3.2 Power (USB 5 V → 3.3 V)

REF	COMPONENT	SPEC / PART	QTY	PURPOSE
J1	USB-C receptacle	16-pin SMD (or Micro-USB)	1	Power + data
R1,R2	Resistor	5.1 kΩ (USB-C CC1/CC2)	2	USB-C sink detection
U2	LDO regulator	RT9013-33 / ME6211C33 (low-dropout, 0.5–1 A)	1	3.3 V rail (low-dropout matters for battery)
C1,C2	Ceramic cap	10 µF / 16 V (LDO in & out)	2	Regulator stability
C3	Bulk cap	22–100 µF on 3V3	1	Absorbs WiFi current spikes
C4–C8	Ceramic cap	100 nF (0402/0603)	5	Per-pin decoupling

3.3 USB-to-UART & auto-reset (programming)

REF	COMPONENT	SPEC / PART	QTY	PURPOSE
U3	USB-UART bridge	CP2102N-A02 or CH340C	1	Serial flashing / monitor
Q1,Q2	NPN transistor	MMBT3904 (SOT-23)	2	DTR/RTS auto-reset into boot mode
R3–R6	Resistor	10 kΩ (EN/IO0 pull-ups, reset net)	4	Strapping + reset circuit
C9	Ceramic cap	1 µF (EN RC delay)	1	Clean power-on reset
D1	ESD array (optional)	USBLC6-2SC6	1	Protect USB D+/D−

3.4 Buttons, I^2C & indicators

REF	COMPONENT	SPEC / PART	QTY	PURPOSE
SW1	Tactile switch	6×3.5 mm SMD	1	EN / Reset
SW2	Tactile switch	6×3.5 mm SMD	1	BOOT (IO0) for manual flashing
R7,R8	Resistor	4.7 kΩ (SDA, SCL pull-ups)	2	I^2C — omit if OLED module already has them
D2	LED + R9 1 kΩ	0603 LED, green	1	Power indicator
D3	LED + R10 1 kΩ (optional)	0603 LED on a free GPIO (e.g. GPIO13)	1	Status / heartbeat
J2	OLED header / FPC	4-pin 2.54 mm or JST-SH	1	Connect/mount the panel
J3	GPIO breakout (optional)	2.54 mm header	1	Future sensors/servos

3.5 Battery / portable option (add if untethered)

REF	COMPONENT	SPEC / PART	QTY	PURPOSE
BT1	LiPo cell	3.7 V, 1000–2000 mAh, JST-PH 2.0	1	Portable power

U4	Charger IC/module	TP4056 + DW01A + FS8205 (charge + protection)	1	Safe USB charging
	Power switch	SPDT slide switch	1	On/off

Battery + LDO: a LiPo runs 3.0–4.2 V. Use a **low-dropout** LDO (RT9013/ME6211, ~250 mV dropout) — an AMS1117 (~1.1 V dropout) would brown-out the ESP32 as the battery drains.

4. Power analysis

LOAD	TYPICAL	PEAK	NOTES
ESP32 — WiFi active	80–160 mA	~250–500 mA (TX bursts)	Spikes need bulk capacitance nearby
SSD1306 OLED	10–20 mA	~27 mA (all pixels on)	Self-emissive; scales with lit pixels
Board total	**~100–180 mA**	**~0.5 A**	Size the LDO for ≥0.5 A (1 A ideal)

- USB 5 V @ 500 mA is plenty for the wired build.
- Battery runtime ≈ capacity ÷ average draw. A 1000 mAh LiPo ≈ **5–8 h** typical (less with constant WiFi traffic).
- Put a 22–100 µF bulk cap + 100 nF right at the module's 3V3/GND to ride out WiFi current spikes (prevents random resets).

5. Design notes & gotchas

- **Flash size:** OTA keeps two app copies — use a module with **≥4 MB** flash and the *Minimal SPIFFS (1.9 MB APP w/ OTA)* partition (already set in your cloud build).
- **I²C pull-ups:** need ~4.7 kΩ on SDA & SCL. Most OLED modules include them — if so, leave R7/R8 unpopulated (two sets in parallel lower the resistance).
- **Antenna keep-out:** the WROOM PCB antenna must overhang the board edge with *no copper/ground pour* beneath it, or WiFi range suffers.
- **Strapping pins:** avoid loading GPIO0, 2, 5, 12, 15 at boot. Your design only uses GPIO21/22 for I²C — safe.
- **Auto-reset:** the CP2102/CH340 DTR+RTS through Q1/Q2 lets the toolchain reset into download mode (no button presses).
- **Decoupling:** one 100 nF as close as possible to each IC power pin.
- **OLED mounting:** if you remove the module's header and solder the bare panel, confirm its pin order (some are GND-VCC-SCL-SDA, others VCC-GND-...).

6. Quick-start shopping list (Path A)

- 1× ESP32-DevKitC (WROOM-32, 4 MB)
- 1× 0.96" SSD1306 128×64 I²C OLED
- 4× female–female jumper wires
- 1× USB cable (matches the board)
- 1× breadboard or perfboard (optional)
- 1× enclosure / 3D-printed face frame (optional)

> This is enough to fully run the project today. Move to Path B only when you want a single integrated board for a finished product.